

Qatar-4b

R-2658

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_181122 · created 2026-08-30T00:01:08+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458444.77333 +/- 0.00069 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.102 +/- 0.052
Transit depth (Rp/Rs)^2	1.03 +/- 1.06 [%]
Orbital Inclination (inc)	82.53 +/- 2.21
Airmass coefficient 1 (a1)	0.945 +/- 0.08
Airmass coefficient 2 (a2)	0.039 +/- 0.036
Scatter in the residuals of the lightcurve fit is	8.5 %
Best Comparison Star	#2 - [421, 140]
Optimal Aperture	2.82
Optimal Annulus	8.03
Transit Duration (day)	0.041 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.102 $\pm$ 0.052 vs 0.138 (deviation 0.69 σ)
Duration fit vs published	0.041 d vs 0.0892 d (deviation 1.61 σ)
Mid-time O-C	0.7 min
Depth SNR	2.0
Residual scatter	8.5 %

Light curve

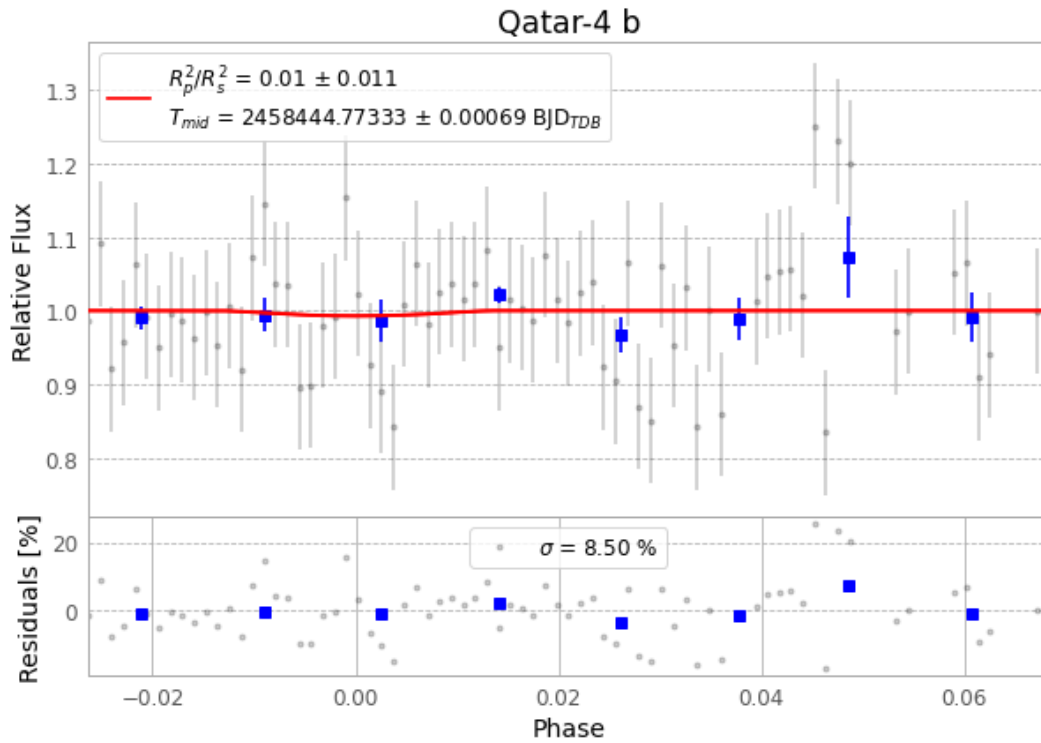


Figure 1 — FinalLightCurve_Qatar-4 b_2018-11-21.png (SHA-256 a6626df418fd19e5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [308, 352], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 2a6a1a6b973e7485...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

83 FITS frames (55 MB), Qatar-4181122051812.FITS → Qatar-4181122092412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-181122.log (08dfb2c457b...), FinalLightCurve_Qatar-4 b_2018-11-21.png (a6626df418fd...), FinalLightCurve_Qatar-4 b_2018-11-21.csv (293ee8dfff86...)

Compute resources

Wall time 138.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 00:01Z.